CHAPTER 3

Conditional Probability and Independence

3.1 Conditional Probability

The concept of conditional probability is useful when we wish to discuss the probability of an event when it is known that some other event had occurred. That is, it provides a method for updating or revising probabilities in light of new information. For example, during the start of the week, a weather forecaster might say the probability of that it will rain over the weekend is 50%, during midweek she might revise this probability to 70% and finally on Friday she might receive even more reliable information and say there is a chance of 60% that it will rain over the weekend. In statistical jargon, we would say each forecast was conditional on the information available up to that point in time.

Definition 3.1.1. Let A and B be events of the sample space S. The conditional probability of event B occurring given that event A has occurred is denoted by $\mathbb{P}\big(B|A\big)$ and defined as

$$\mathbb{P}\big(B|A\big) = \frac{\mathbb{P}\big(A \cap B\big)}{\mathbb{P}\big(A\big)} \tag{3.1}$$

Example 3.1.1. A woman with a medical background is one of the 198 applicants for an MBA programme in which 81 will be selected. Suppose the lady hears, along the grapevine, on good authority, that there were 70 woman applicants, of whom 38 were selected. Assess the probabilities of being accepted before and after the lady received the grapevine information. (Use the definition of conditional probability.)

Solution 3.1.1. Clearly, the probability of being accepted into the MBA before the grapevine information is $81/198 = 0.409$ and after the information, the probability of acceptance became $38/70 = 0.543$.

It can be noted that initially the probability of being accepted is just the number of openings divided by the total number of applicants. Now in light of the new evidence, the probability is revised to consider only the female applicants, of which 38 were selected from a total of 70 applicants.

Example 3.1.2. Suppose A and B are two events in a sample space S and that $\mathbb{P}(A) = 0.6$, $\mathbb{P}(B) = 0.2$ and $\mathbb{P}(A|B) = 0.5$. Find the following probabilities

(a) $\mathbb{P}(B|A)$

(b) $\mathbb{P}(A \cup B)$

(c) $\mathbb{P}(\bar{A} \cap B)$

(d) $\mathbb{P}(B|\bar{A})$

Solution 3.1.2.

(a) By definition of conditional probability $\mathbb{P}(A|B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, so that from the given information we have

$$0.5 = \frac{\mathbb{P}(A \cap B)}{0.2}$$

Thus, $\mathbb{P}(A \cap B) = 0.5(0.2) = 0.10$. Similarly,

$$\begin{aligned}\mathbb{P}(B|A) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \\ &= \frac{0.10}{0.60} = 1/6\end{aligned}$$

(b) It is straightforward that

$$\begin{aligned}\mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \\ &= 0.6 + 0.2 - 0.1 = 0.7.\end{aligned}$$

(c) Recall that $B = (B \cap A) \cup (B \cap \bar{A})$, and these two are mutually exclusive. Therefore,

$$\begin{aligned}\mathbb{P}(\bar{A} \cap B) &= \mathbb{P}(B) - \mathbb{P}(A \cap B) \\ &= 0.2 - 0.1 \\ &= 0.1.\end{aligned}$$

(d) From (c) and definition of conditional probability, we have that

$$\begin{aligned}\mathbb{P}(B|\bar{A}) &= \frac{\mathbb{P}(B \cap \bar{A})}{\mathbb{P}(\bar{A})} \\ &= \frac{0.1}{1 - 0.6} \\ &= 0.25.\end{aligned}$$

Example 3.1.3. Is it possible for events A and B in a sample space to have the following probabilities $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B) = 0.8$ and $\mathbb{P}(A|B) = 0.2$

Solution 3.1.3. Lets first find $\mathbb{P}(A \cap B)$ so that we can evaluate $\mathbb{P}(A \cup B)$. From the definition of conditional probability

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}(A|B)\mathbb{P}(B) \\ &= 0.2(0.8) = 0.16\end{aligned}$$

Thus

$$\begin{aligned}\mathbb{P}(A \cup B) &= 0.5 + 0.8 - 0.16 \\ &= 1.14(> 1)\end{aligned}$$

Hence its impossible to have events A and B in the same sample space as defined above.

Example 3.1.4. Show that $\mathbb{P}(B|A) + \mathbb{P}(\bar{B}|A) = 1$.

Solution 3.1.4. Left as an exercise!

Example 3.1.5. The probability that a first year student passes a Programming module if she passes Statistics is 0.5. The probability that she passes Statistics if she passes Programming is 0.8, and the probability that she passes Statistics is 0.7. The Statistics results come out first, and the student finds she has failed. What is now the conditional probability of passing Programming module?

Solution 3.1.5. First lets define events A and B as follows:

A: student fails Programming

B: student fails Statistics

Then in mathematical notation, we have the following: $\mathbb{P}(A|B) = 0.5$, $\mathbb{P}(B|A) = 0.8$ and $\mathbb{P}(B) = 0.7$. Now we need to evaluate $\mathbb{P}(A|\bar{B})$. That is,

$$\mathbb{P}(A|\bar{B}) = \frac{\mathbb{P}(A \cap \bar{B})}{\mathbb{P}(\bar{B})} = \frac{\mathbb{P}(A) - \mathbb{P}(A \cap B)}{\mathbb{P}(\bar{B})}$$

Now we need to evaluate $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \cap B)$. From the definition of conditional probability

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}(A|B)\mathbb{P}(B) \\ &= 0.5(0.7) = 0.35\end{aligned}$$

Also from the definition of conditional probability, we can write

$$\begin{aligned}\mathbb{P}(A) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B|A)} \\ &= 0.35/0.8 = 0.4375.\end{aligned}$$

Hence

$$\begin{aligned}\mathbb{P}(A|\bar{B}) &= \frac{0.4375 - 0.35}{1 - 0.7} = \frac{0.0875}{0.3} \\ &= 0.292.\end{aligned}$$

Note that the definition of $\mathbb{P}(B|A)$ is consistent with the interpretation of probability as a limiting frequency. For instance, suppose an experiment is repeated for N times, $N \to \infty$. Then if we consider only experiments in which A occurs, then $\mathbb{P}(B|A)$ will be the long-run relative frequency in which B occurs. Therefore conditional probability also respects the axioms of probability.

Proposition 3.1.1. Let A and B be events in the sample space, S. Then

AI. $0 \leq \mathbb{P}(B|A) \leq 1$.

AII. $\mathbb{P}(S|A) = 1$.

AIII. Also let B_1 and B_2 be events in S. If B_1 and B_2 are mutually exclusive then

$$\mathbb{P}\big(B_1 \cup B_2 \big| A\big) = \mathbb{P}(B_1|A) + \mathbb{P}(B_2|A)$$

Proof. Left as an exercise for the students.

Multiplication rule.

Let $A_1, A_2, \ldots, A_n$ be events of the sample space S. Then

$$\mathbb{P}\big(A_1 \cap A_2 \cap \ldots \cap A_n\big) = \mathbb{P}\big(A_1\big)\mathbb{P}\big(A_2|A_1\big)\mathbb{P}\big(A_3|A_2, A_1\big)\ldots\mathbb{P}\big(A_n|A_{n-1}, \ldots, A_2, A_1\big)$$

The multiplication rule is proved by simply applying the conditional probability.

3.2 Bayes' Theorem

Let A and B be events of a sample space S. Then we can express B as

$$B = (B \cap A) \cup (B \cap \bar{A}).$$

Since $B \cap A$ and $B \cap \bar{A}$ are mutually exclusive, we have that

$$\mathbb{P}(B) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap \bar{A})$$

Thus, from definition of conditional probability

$$\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|\bar{A})\mathbb{P}(\bar{A}) \tag{3.2}$$

That is probability of the event B occurring is an weighted average of the conditional probability given A has occurred and conditional probability of B given that A has not occurred. This is very useful formula especially in instances where its difficult to compute the probability of an event directly but easier to compute it once we know whether or not some other event has occurred.

Example 3.2.1. An insurance company believes that people can be divided into two classes – accident-prone and those who are not. The company statistics show that an accident-prone person will have an accident at some point within the fixed 12-month period with probability of 0.4, whereas this probability decreases to 0.2 for a person who is not accident-prone. If we assume that 30% of the population is accident-prone, what is the probability that a new policy-holder will have an accident within a year of purchasing the insurance policy?

Solution 3.2.1. Let AP represent an accident-prone individual. Then from the statement above, we are told 30% of the population is accident-prone, i.e.

$$\mathbb{P}(AP) = 0.3$$

Also let A be the event that an individual will have an accident within the first 12 months after taking the policy. Then we have that

$$\mathbb{P}(A|AP) = 0.4$$

and

$$\mathbb{P}(A|\overline{AP}) = 0.2$$

Therefore by equation 3.2 we have that

$$\begin{aligned}\mathbb{P}(A) &= 0.4(0.3) + 0.2(0.7)\\ &= 0.26\end{aligned}$$

Example 3.2.2. Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that s/he is accident?

Solution 3.2.2. Note that this question now is asking for the reverse conditional probability. That is, the probability of being accident prone given that you had an accident within 12 months from purchasing a policy.

$$\mathbb{P}(AP|P) = \frac{\mathbb{P}(AP \cap A)}{\mathbb{P}(A)}$$

$\mathbb{P}(A)$ was computed in the example above and $\mathbb{P}(AP \cup A)$ can be evaluate from $\mathbb{P}(AP)$ and $\mathbb{P}(A|AP)$. That is,

$$\begin{aligned}\mathbb{P}(AP|P) &= \frac{0.4(0.3)}{0.26} \\ &= 6/13.\end{aligned}$$

In the example above, we shown how one can find $\mathbb{P}(A|B)$ from the probabilities of $\mathbb{P}(B|A)$ and $\mathbb{P}(B|\bar{A})$ and probability of $\mathbb{P}(A)$. This relationship is what we refer to as Bayes' theorem.

Theorem 3.2.1 (Bayes' Theorem)**.** Let A and B be events of the sample space, S. Then

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|\bar{A})\mathbb{P}(\bar{A})} \tag{3.3}$$

Proof. Let A and B be some events in S. Then by definition of conditional probability, we have that

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Still from the definition of conditional probability, we can write $\mathbb{P}(A \cap B)$ as

$$\mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A)$$

We also have from 3.2 that

$$\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|\bar{A})\mathbb{P}(\bar{A})$$

Hence

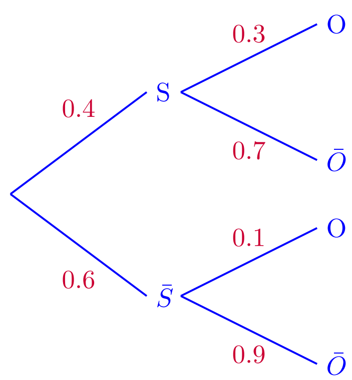
$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|\bar{A})\mathbb{P}(\bar{A})}$$

Example 3.2.3. A well is drilled as part of an oil exploration programme. The probability of the well passing through shale is 0.4. If the well passes through shale, the probability of striking oil is 0.3. If it does not pass through shale, it drops to 0.1.

(a) Given that oil was found, what is the probability that it did not pass through shale?

(b) Given that oil was not found, what is the probability that it passed through shale?

Solution 3.2.3. Let `S` represent the event that a drilled well passes through the shale and `O` be event that oil is found. Then the tree diagram representing the information presented above is given as follows:



(a) We are asked to find $\mathbb{P}(S|O)$ and by Bayes theorem:

$$\begin{aligned}\mathbb{P}(S|O) &= \frac{\mathbb{P}(O|\bar{S})\mathbb{P}(\bar{S})}{\mathbb{P}(O|S)\mathbb{P}(S)+\mathbb{P}(O|\bar{S})\mathbb{P}(\bar{S})}\\ &= \frac{0.1(0.6)}{0.3(0.4)+0.1(0.6)}\\ &= \frac{1}{3} = 0.333.\end{aligned}$$

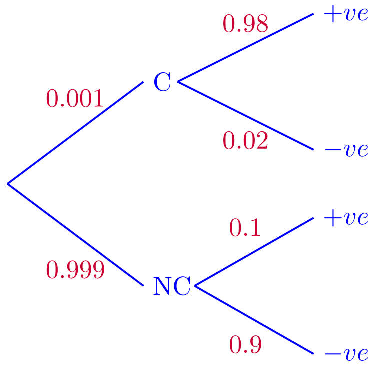
(b) Here, we are to find $\mathbb{P}(S|\bar{O})$ and analogous to the answer given above:

$$\begin{aligned}\mathbb{P}(S|\bar{O}) &= \frac{0.7(0.4)}{0.7(0.4)+0.9(0.6)}\\ &= \frac{14}{41} = 0.341.\end{aligned}$$

Example 3.2.4. The probability that a cancer test will detect the disease in a person who has cancer is `0.98`. The probability that a person who does not have cancer will give a positive reading in the test is `0.1`. (i.e. the test says he has the disease even though he has not). If `0.1%` of the population has cancer, what is the probability that a person selected at random will in fact have cancer, given that he shows a positive reading on the cancer test?

Solution 3.2.4. The tree diagram representing the information presented above is given in the figure below and from this figure and Bayes theorem we have that:

$$\begin{aligned}\mathbb{P}(C|+ve) &= \frac{\mathbb{P}(+ve|C)\mathbb{P}(C)}{\mathbb{P}(+ve|C)\mathbb{P}(C)+\mathbb{P}(+ve|NC)\mathbb{P}(NC)}\\ &= \frac{0.98(0.001)}{0.98(0.001)+0.1(0.999)}\\ &= 0.00971.\end{aligned}$$



Law of Total Probability

Let $A_1, A_2, \ldots, A_n$ be a set of pairwise mutually exclusive and exhaustive events of the sample space, S. Let B be any other event in S. Then

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B|A_i)\mathbb{P}(A_i) \tag{3.4}$$

Proof. Events A_i for $i = 1, 2, \ldots, n$ are said to be exhaustive events which implies that

$$S = A_1 \cup A_2 \cup \ldots \cup A_n$$

We also know that

$$B = B \cap S = B \cap \big(A_1 \cup A_2 \cup \ldots \cup A_n\big)$$

Thus,

$$B = (B \cap A_1) \cup (B \cap A_2) \cup \ldots \cup (B \cap A_n)$$

Since the A_i's are pairwise mutually exclusive then it follows that $B \cap A_i$ for $i = 1, 2, \ldots, n$ are also mutually exclusive. Therefore

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}\Big[(B \cap A_1) \cup (B \cap A_2) \cup \ldots \cup (B \cap A_n)\Big] \\ &= \mathbb{P}(B \cap A_1) + \mathbb{P}(B \cap A_2) + \ldots + \mathbb{P}(B \cap A_n)\end{aligned}$$

Written compactly as

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B|A_i)\mathbb{P}(A_i)$$

□

General Bayes Theorem

From the law of total probability, we can then generalize the Bayes theorem as follows:

$$\mathbb{P}(A_k \mid B) = \frac{\mathbb{P}(B \mid A_k)\mathbb{P}(A_k)}{\sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)} \tag{3.5}$$

where A_i for $i = 1, 2, \ldots, n$ are pairwise mutually exclusive and exhaustive events and B any event defined on the sample space, S.

3.3 Independent Events

If the occurrence of event A has nothing to do with the occurrence B. Then we expect the conditional probability of B given that the event A has occurred to be the same as the unconditional probability of B:

$$\mathbb{P}(B \mid A) = \mathbb{P}(B).$$

That is, the information that event A has occurred does not change the probability of B occurring. If that's the case, then using the definition of conditional probability, we have that

$$\frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \mathbb{P}(B)$$

i.e.

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Definition 3.3.1. Events A and B are said to be *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \tag{3.6}$$

Example 3.3.1. The probability that Botswana Pula (BWP) will gain against the South African Rand (ZAR) is 0.53. The probability that you will wake up late tomorrow is 0.42. What is the probability that tomorrow, the BWP weakens against the ZAR and you wake up late?

Solution 3.3.1. Let A be the event that BWP gains against the ZAR and B be waking up late. It is safe to assume that these two events are independent and therefore

$$\begin{aligned}\mathbb{P}(\bar{A} \cap B) &= \mathbb{P}(\bar{A})\mathbb{P}(B) \\ &= (1 - 0.53)(0.42) = 0.1974.\end{aligned}$$

Example 3.3.2. A satellite is to have a number of solar panels, which will function independently of each other, and each will fail during the mission with probability 0.05. For success, at least one must function until the end of the mission. How many solar panels are necessary, if the probability of success must exceed 0.999?

Solution 3.3.2.

$$\mathbb{P}(\text{ at least one success }) = 1 - \mathbb{P}(\text{ no success })$$

Thus, we need to find the probability that all the panels will fail given that the operate independently. That probability for n panels is 0.05^n. Therefore, it is required that

$$1 - 0.05^n \geq 0.999$$

for the mission to be deemed successfull. That is,

$$0.05^n \leq 0.001$$

After simple algebraic computations, we get that

$$n \geq 2.3058$$

Hence $n = 3$ panels will be needed to 99.9% certain that the mission will be a success.